

A WATERMARKING SEQUENCE USING PARITIES OF ERROR CONTROL CODING FOR IMAGE AUTHENTICATION AND CORRECTION

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Abstract

A novel function for image watermarking is proposed. We show how watermarking sequence can be used for correction of illegal modifications. To make this task possible, the parities generated by the conventional error control coding (ECC) technique is used for the watermarking sequence. Then, the receiver can correct any alterations by applying the ECC decoding. To increase the correction capability we also adopt a scrambling scheme before the encoding to change a burst type of content modifications (i.e., errors) into a random noise. The scrambling key can be also used as a key for the authentication of the sender.

Keywords: watermarking, authentication, ECC, RS codes

I. INTRODUCTION

To protect the ownership and to guarantee the secure delivery of images, it is necessary to embed watermarks to digital images. To date, three different functions for digital watermarks have been investigated [1]. In the first function, the ownership of the image is the main concern. Here, a watermark is inserted into the image to prove the ownership of the image. In this case, a robust invisible watermark which must be difficult (or impossible) to remove is required [2]. The second function of the digital watermarking is to track the illegal distribution of the images. This is closely related to the ownership function in the sense that the embedded watermarks should be robust to the attacks. However, the main purpose of the watermark of this function is to identify illegal distributors of images rather than to verify its ownership. To achieve this goal, different

watermarks are assigned to different distributors to be able to track the distribution of the image. The third function of the watermarking is to protect the receiver. That is, the watermark is embedded for the receiver to identify the sender and to ensure the content of the image. For the authentication, the watermark should be fragile to make it easy to identify any change in the image [1][3]. There are relatively little works done for this function using watermarking techniques. In the previous methods, only a binary answer can be provided as to whether the image has been altered or not [3][4], or, at most, the locations of the modifications are identified [1]. In [5], a tamper detection method was proposed to distinguish malicious changes such as replacing/adding features from non-malicious changes resulting from common image processing operations.

In this paper, we expand the capability of the watermarking from the upper-mentioned detection and localization of the alterations to the correction. More specifically, we adopt an error control coding (ECC) technique to generate the watermark, which in turn makes it possible to correct and detect the changes from the extracted watermarks. Also, using the scrambling key, we can certify the ID of the sender. Another advantage of the proposed watermarking is that it does not require the original watermark for the verification and correction.

II. PROPOSED WATERMARKING

In this section, we describe the proposed watermarking technique. Based on the error control coding technique, it can detect and correct the alterations of the original image.

A. Watermark Encoding

Fig 1 shows the procedure of the proposed watermarking insertion. At the first step, the lowest significant bits (LSBs) in all pixels of the original image I_o are set to zero. The image provider can choose how many LSBs to be zero depending on the quality of the image distributed. In general, it is reported that the changes of one or two LSBs are not noticeable by the human eyes. Now, we scramble this zero replaced image I_z to obtain $S_K(I_z)$ by a random sequence generated by a key K . The purpose of the scrambling is to make the content modification, which is normally a burst type of alteration, to be a random type error. As a result, the scrambling increases the correction capability of error control coding (ECC). Note also that the key K should be securely transmitted to the receiver. Here, we adopt the public key technique to give the authentication (i.e., to ensure the sender of the image). That is, we encrypt the key K with the sender's private key K_{pri} and send it to the receiver. Then, at the receiver, the public key K_{pub} of the expected sender is selected for the decoding of the key. If the pirate sends the message, then the decoded key will not be identical to the original key K . In this case, we will detect lots of errors, which exceed the correction capability. Therefore, there are two cases for the "change detection without correction" in Fig 2. First, the modified area of the watermarked image is too large and exceeds the correction capability. Second, the decrypted key is not identical to the encryption key. In the second case, the sender of the image may not be the expected person of the receiver. That is, the authentication of the sender is not assured.

With the scrambled image data we generate a parity sequence $W(S_K(I_z))$ by using ECC, which is used as a watermark sequence. After making the watermark sequence, we put it in the place of zeros in LSBs of the zero-padded original image I_z in a raster scan order. Then the watermarked image I_w is ready to be distributed.

B. Authentication and correction of the image

Since the receiver of the watermarked image I'_w does not know if some pirates modify the image I_w or not, we must verify the embedded watermark. As shown in Fig 2, we first need to recover the parities embedded in I'_w . By the raster scan order, the receiver can extract the watermark sequence

$W'(S_K(I_z))$ (ECC parities) from LSBs of I'_w . Second, the receiver sets all LSBs to zero in order to make I'_z and they are scrambled by the same scrambling order with the decrypted key K . It is referred to as $S_K(I'_z)$. Then, $S_K(I'_z)$ is encoded by the same ECC to generate parities $W(S_K(I'_z))$. Then two sequences $W'(S_K(I_z))$ and $W(S_K(I'_z))$ are compared to see if they are same. If the two sequences are matched exactly, then the received image is authenticated and one can sure that there is no change. Otherwise, someone modified it during the transmission of the image or someone else sends the image. When the two sequences are not coincident, it is possible to recover the distributed image I_z by ECC decoding with recovered parities $W'(S_K(I_z))$ and $S_K(I'_z)$ if the amount of the modifications is within the limit of the correction capability of the ECC. Even if it is not correctable because the image is modified too much, i.e., over the limit of the correction capability of the ECC decoding, we can still localize the place of the modification of the image by comparing the different positions of the sequence.

The image provider and the legal receiver do not want that the transmitted image is modified during the transmission, and they may want to recover the original image transmitted even when it is altered because the retransmission of the image wastes time and resource. Hence, the ECC is adopted to generate the watermark sequence to endow the watermarking scheme with the correction capability.

C. ECC for the watermark sequence

There are some error control coding schemes and every method generates parities. The generated parities can be used for the watermark sequence. In our experiment, we exploit the product coding of Reed-Solomon (RS) [6] codes since we can assume that pirates change the image in a form of block. RS code shows the distinguished performance for burst errors. The structure of product coding is given in Fig 3. It produces parities on rows, parities on columns and parities on parities. For a $K_1 \times K_2$ image, we first generate the RS code words for each row. That is, for each row with K_2 bytes, we generate the RS code words with P_1 parity bytes. After generating RS code words for all rows of the image, we then apply the RS encoding for each column of the image including the P_1 parity columns, generated from the row execution, to obtain P_2 parity bytes for each column. Then, the

total number of parity bytes is $P_1K_1+P_2K_2+P_1P_2$ and they are inserted into the distributed image as a watermark to be used for the verification and correction.

III. EXPERIMENTAL RESULTS

To show the performance of the proposed watermarking scheme, we applied the watermarking technique to real images. The watermark sequence (i.e., ECC parities) is inserted to two LSBs for each gray level. Therefore, before the ECC code words generation, we need to replace two LSBs of each gray level with zeros and to scramble them with a randomly generated order. Then, for those zero inserted and scrambled image data, we calculate the ECC code words to generate the watermark. Now, the generated watermark is inserted to two LSBs of the gray levels by visiting the pixels in the raster scan order. Since we insert the parity bytes to the two LSBs of the image pixels, for 8 bits/pixel black-and-white image, it can accept $K_1K_2/4$ bytes of the parities. So, positive integers P_1 and P_2 should satisfy

$$\frac{K_1K_2}{4} \geq P_1K_1 + P_2K_2 + P_1P_2. \quad (1)$$

Note that if the RS codes have P_1 and P_2 parities, then the maximum number of correctable error bytes are $P_1/2$ and $P_2/2$, respectively. Thus, if the image contains important information in row(column)-wise pattern, then we have to emphasize the column(row)-wise correction capability because it is reasonable to assume that the pirate modify the row(column)-wise information block. We assume that rows and columns are equally important. Therefore, we use the same number of parity bytes, i.e., $P_1 = P_2 = P$. Also, we assume that the lengths of row and column are equal, i.e., $K_1 = K_2 = K$. Furthermore, since RS code is a q^m -ary BCH code of length q^m-1 , and one pixel is represented by 8 bits, the length of the code word is equal to $255(=2^8-1)$. With these assumptions, the optimal image size for the correction capability becomes 229×229 as shown in the Table 1. Thus, we can correct 13 ($P/2$) bytes in a row of image and a column of that. If the actual image size is larger than the optimal image size, then we insert the parities only into the inside of the image that covers the optimal size. Or, to approximately match the actual image size and the

optimal one, we can control the size of the parity byte P .

To show the performance of the proposed algorithm, we applied it to real images. Two of the results are shown in Fig 4 and 5. For the original image Fig 4,5-(a), we insert the RS parities (i.e., the watermark) to the image data as in Fig 4,5-(b). Then, the watermarked images are altered by inserting a cat (Fig 4,5-(c)) in the watermarked images. At the receiver, after the watermark extraction and correction, we can correct the alteration and restore the original images completely as shown in Fig 4,5-(d).

IV. CONCLUSIONS

In this paper, we have proposed a novel watermarking method using the error control coding technique. The proposed method can correct alterations in the watermarked images. As a result, it creates a new functionality for the watermarking technique, namely the correction of alterations. Another advantage of the proposed watermarking is that the original watermark is not required for the verification and correction.

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Table 1. Relationships between parameters when $N=255$ (unit is byte.)

K	P	$K^2/4$	P^2+2PK
227	28	12882	13496
229	26	13110	12584
231	24	13340	11664
233	22	13572	10736

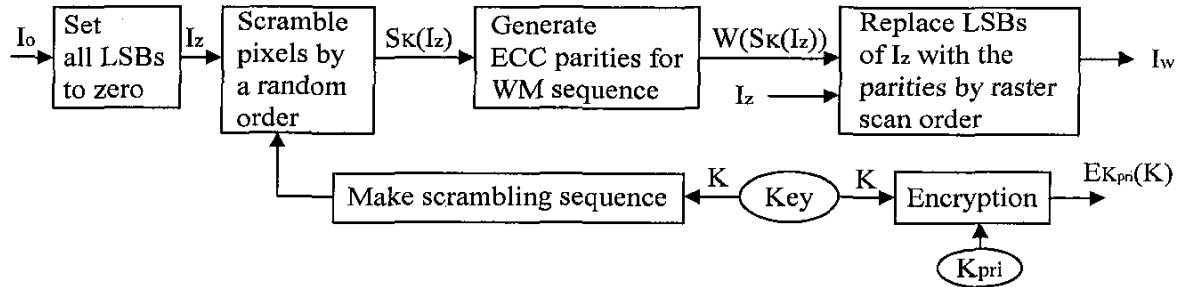


Fig. 1 Embedding of invisible image watermarking

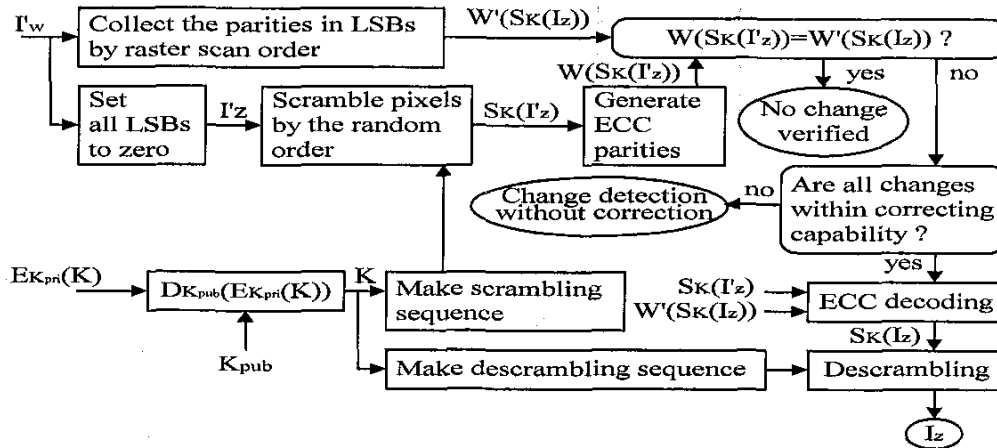


Fig. 2 Watermark extraction and verification

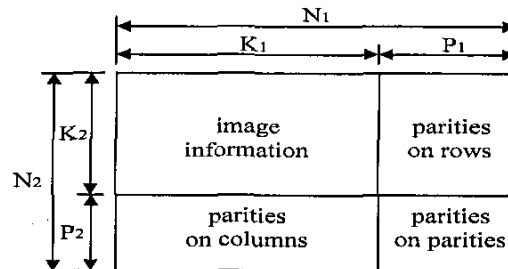


Fig. 3 The structure of product coding.

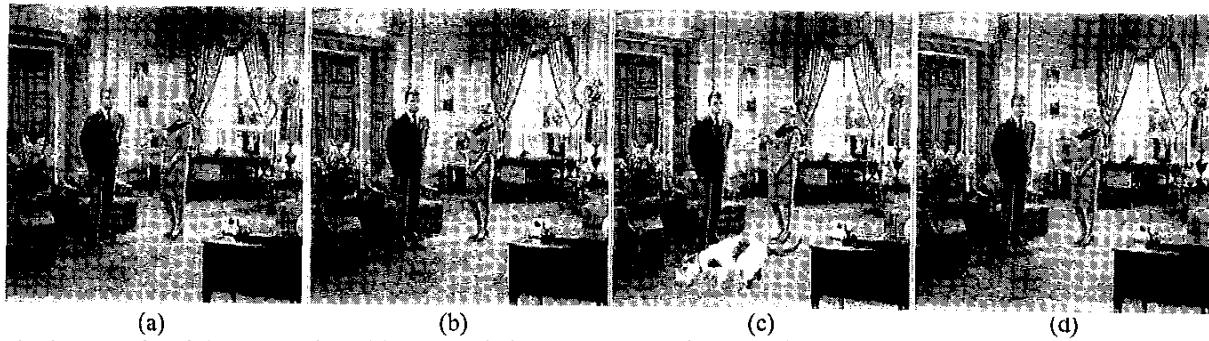


Fig.4 Example of the correction. (a) Original, (b) Watermarked image, (c) Altered image, (d) Corrected image.

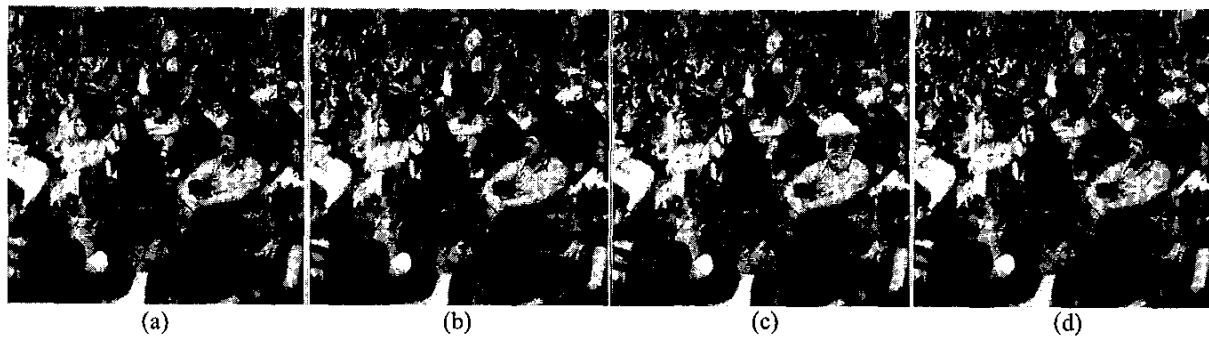


Fig.5 Example of the correction. (a) Original, (b) Watermarked image, (c) Altered image, (d) Corrected image.

Biography

Jaejin Lee received the B.S. degree from Yonsei University, Seoul, Korea, in 1983. He received the M.S.E.E. degree from the University of Michigan, Ann Arbor, MI, in 1984 and the Ph.D. degree from the Georgia Institute of Technology, Atlanta, GA, in 1994. In 1997, he joined the faculty at Dongguk University, Seoul, Korea, and he is currently an Assistant Professor of Electronic Engineering. His research interests are coding for storage systems and watermarking for image.

Chee Sun Won received his BS degree in electronics engineering from Korea University, Seoul, in 1982 and MS and PhD degrees in electrical and computer engineering from the University of Massachusetts at Amherst in 1986 and 1990, respectively. From 1989 to 1992 he was a senior engineer with GoldStar Co., Ltd., in Seoul. In 1992, he joined Dongguk University, where he is currently an associate professor in the Electronics Engineering Department. His research interests include image segmentation, digital video coding, digital video browsing and retrieval, and video securities.